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| **Grade** | **15.00** out of 15.00 (**100**%) |

Question **1**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It only works on small datasets
c. It allows for efficient parallel processing and sorting of data
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **2**

Complete

Mark 1.00 out of 1.00

**The process of placing or rearranging a collection of elements into a particular order is known as**

a. Merging
b. Rearranging
c. Searching
d. Sorting

The correct answer is: Sorting

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort

b. Heap sort

c. Merge sort

d. Quick sort

The correct answer is:
Merge sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary

b. It divides the list into two halves, sorts each half, and then merges them

c. It builds a sorted array one element at a time

d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort

b. Insertion Sort

c. Bubble Sort

d. Merge Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Heap Sort
- c. Quick Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Bubble Sort
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It complicates the analysis process
- c. It makes the analysis slower
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **9**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By comparing elements sequentially
- b. By using the bubble-up method
- c. By sorting data in a single pass
- d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes code execution slower
- b. Makes data harder to manage
- c. Decreases the efficiency of algorithms
- d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **11**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 40,80,90,150,200,300,400,470,70,120
- c. 200,470,80,150,40,90,300,400,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It is less efficient than custom sorting algorithms
c. It only works with integer arrays
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It bubbles up the largest element in each pass
c. It recursively sorts subproblems
d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Insertion Sort
b. Quick Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Sorting the entire list sequentially
- c. Combining sorted sublists
- d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems